Computer Science III	Lab C – LIST FUNHOUSE, Instance Variable Version

Lab Goal :

This lab was designed to teach you more about linked lists. In this lab the list is stored in a `ListInstanceVarFunHouse`, unlike last lab where the list was created in the main method and passed in as a parameter to the static methods.

Lab Description :

Use the `ListNode` class given below to write some basic Linked List methods. In this implementation only a reference to the first node is stored in a reference variable. You access the other blocks by iterating to them from the first node.

ListNode – stores a value and a reference to the next node

```
public class ListNode implements Linkable
{
   private Comparable listNodeValue;
   private ListNode nextListNode;

   public ListNode(){
      listNodeValue = null;
      nextListNode = null;
   }

   public ListNode(Comparable value, ListNode next){
      listNodeValue=value;
      nextListNode=next;
   }

   public Comparable getValue(){
      return listNodeValue;
   }

   public ListNode getNext(){
      return nextListNode;
   }

   public void setValue(Comparable value){
      listNodeValue = value;
   }

   public void setNext(Linkable next){
      nextListNode = (ListNode)next;
   }
}
```

Sample Data :

See the main of lab15c.

Sample Output :

```
                         Original list values:  go on at 34 2.1 -a-2-1 up over
                                                 num nodes = 8
        List values after calling nodeCount:  go on at 34 2.1 -a-2-1 up over
      List values after calling doubleFirst:  go go on at 34 2.1 -a-2-1 up over
       List values after calling doubleLast:  go go on at 34 2.1 -a-2-1 up over over
   List values after calling skipEveryOther:  go on 34 -a-2-1 over
 List values after calling removeXthNode(2):  go 34 over
List values after calling setXthNode(2,one):  go one over
 List values after calling removeXthNode(1):
```